

SEMESTER END EXAMINATION (SEE) B. E. 5th SEMESTER – DEC- 2013

11CS303 – Compiler Design

Time: 3 Hrs	Answer All Questions	Max Marks: 100
1. a)	Explain the structure of a compiler with neat diagram. Discuss first four stages in detail.	2+8=10
b)	What is meant by input buffering? Explain with an algorithm.	6
c)	Write a LEX program to count the given no. of comment lines in a given C program. Also eliminate them and copy to a separate file. Provide input and output file names using command line arguments.	4
2. a)	Write regular definition to recognize the following:	4
	i) fax number of form : 910-(202)-777-8888	
	ii) Email address of form: <u>abc@vtu.ac.in</u>	
b)	Write down the algorithm for finding First and Follow Set. Compute First and Follow set for the given grammar applying above algorithms.	2+2+2+4=10
	$S \rightarrow AB$ $A \rightarrow Ca \epsilon$ $B \rightarrow BaAC c$ $C \rightarrow b \epsilon$	
c)	What do you mean by LL(1) grammar? Write down the conditions for the same. Eliminate the left recursions for the following grammars	6
	(a) $S \rightarrow S(S)S c$ (b) $S \rightarrow SS+ SS^* a$ (c) $S \rightarrow Aa b, A \rightarrow Ac Sd c$	
3. a)	Construct LR(0) automaton for the given expression grammar and discuss the steps required to construct the same.	6+4+4=14
	$E \rightarrow E+T T$ $T \rightarrow T * F F$ $F \rightarrow (E) id.$	
	Also construct the parsing table for same. Also write down the moves for the input id*id+id (Parse this input and check whether it is accepted by the grammar).	
b)	What is shift reduce parser? Explain how shift reduce parser will parse the input id1*id2 using the grammar given in Q. 3(a). Discuss the conflicts that occur during shift reduce parsing.	6
4. a)	What do you mean by syntax directed definition (SDD) and translation? Explain with example attributes of a grammar. How do you represent SDD for LL(1) and LR grammar. Give the annotated parse tree for the following expression: (3+4)*(3+6) for the given grammar	7
	$E \rightarrow E+T T$ $T \rightarrow T * F F$ $F \rightarrow id / number / (E)$ $id \rightarrow [a_zA-Z]^+$ $number \rightarrow [0-9]^+$	